NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-4(P1) | Date: 09-04-26
Time: 3 Hrs | 2024_P1 | Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**A**	2	**B**	3	**A**	4	**A**	5	**ABD**
6	**A**	7	**AD**	8	**152**	9	**6**	10	**6**
11	**1**	12	**229**	13	**23**	14	**C**	15	**A**
16	**A**	17	**A**						

PHYSICS

18	**D**	19	**D**	20	**C**	21	**B**	22	**AB**
23	**ABD**	24	**BC**	25	**6**	26	**15**	27	**28**
28	**48**	29	**634**	30	**7**	31	**D**	32	**A**
33	**B**	34	**C**						

CHEMISTRY

35	**B**	36	**C**	37	**A**	38	**A**	39	**AB**
40	**CD**	41	**AC**	42	**9**	43	**2**	44	**18**
45	**4**	46	**25**	47	**4**	48	**D**	49	**A**
50	**C**	51	**B**						

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SOLUTIONS
MATHS

1.

$$P_n = \frac{1}{6}\left\{\frac{5}{6}\cdot\frac{1}{6}\cdot P_{n-3} + \frac{5}{6}\cdot P_{n-2}\right\} + P_{n-1}\cdot\frac{5}{6}$$

$$216P_n - 5P_{n-3} - 30P_{n-2} - 180P_{n-1} = 0$$

2. We can write as

$$x = \sqrt[3]{\frac{4}{25}} = \left(\frac{4}{25}\right)^{1/3} = \left(\frac{2}{5}\right)^{2/3}.$$

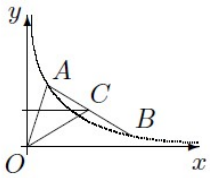
The equation $x^x = y^y$ is equivalent to $x \log x = y \log y$. We can rewrite $x \log x$ as

$$\begin{aligned} x \log x &= \left(\frac{2}{5}\right)^{2/3} \cdot \frac{2}{3} \log \frac{2}{5} \\ &= \left(\frac{2}{5}\right)^{2/3} \cdot \frac{2}{5} \cdot \frac{5}{3} \log \frac{2}{5} \\ &= \left(\frac{2}{5}\right)^{5/3} \cdot \frac{5}{3} \log \frac{2}{5} \\ &= \left(\frac{2}{5}\right)^{5/3} \log \left(\frac{2}{5}\right)^{5/3}. \end{aligned}$$

By inspecting the right side, we see that $y = \left(\frac{2}{5}\right)^{5/3}$ s
$x \log x = y \log y$. Furthermore $0 < y < x$. So we have fo

$$y = \left(\frac{2}{5}\right)^{5/3} = \left(\frac{2^5}{5^5}\right)^{1/3} = \left(\frac{32}{3125}\right)^{1/3} = \boxed{\sqrt[3]{\frac{32}{3125}}}.$$

3. We have $A\left(x_1, \frac{1}{x_1}\right), B\left(x_2, \frac{1}{x_2}\right)$ and $C\left(\frac{x_1+x_2}{2}, \frac{1}{2x_1}+\frac{1}{2x_2}\right)$. Computing the coordinates of $\bar{v} = |OC|.\bar{AC} + |AC|.\bar{OC}$ we find that the vector $\bar{v}$ and so also bisector of the angle $\angle OCA$ is parallel to x axis. Since $|OA| = |AC|$ this yields $\angle AOC = \angle ACO = 2\,.\,\angle COx$ (see figure 3) and $\angle AOX = \angle AOC + \angle COX = 3.\angle COX$



4. Let S_n denote the number of ways that n women in n seats can be reseated so that each woman reseats herself in the seat she occupied before or a seat next to it. Clearly $S_1 = 1$ and $S_2 = 2$

consider the case with $n \geq 3$ women, and focus on the woman at the right end of the line. If this woman sits again in this end seat, then the remaining $n-1$ women can reseat themselves in S_{n-1} ways. If this end woman sits in the seat next to hers, then the former occupant of this new seat must sit on the end. Then the remaining $n-2$ women can seat themselves in S_{n-2} ways. Thus for $n \geq 3$, $S_n = S_{n-1} + S_{n-2}$. Therefore

Computing S_1, S_2, S_3..... we have $S_{10} = 89$

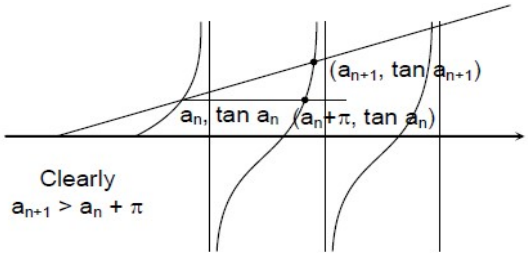
5.

$$\frac{dy}{dx} = \cos x\Big|_{x=a_n} = \frac{\sin a_n}{a_n + \frac{\pi}{2}}$$

$$a_n + \frac{\pi}{2} = \tan a_n$$

Taking intersection of y = tan x

and $y = x + \frac{\pi}{2}$



Clearly

$a_{n+1} > a_n + \pi$

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6. Let $z = x + iy$

A and B are real and imaginary part of $z^2 = z^{-1} + 3i$

Similarly C and D are real and imaginary part of $z^3 - 3iz = 1$

$\Rightarrow (x, y) \in A \cap B \Leftrightarrow (x, y) \in C \cap D \Rightarrow A \cap B = C \cap D$

7. *Let* $P = A + \omega B$ where $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$. We have

$P\bar{P} = (A + \omega B)(A + \bar{\omega} B) = A^2 + \omega BA + \bar{\omega} AB + B^2$

$= AB + \omega BA + \bar{\omega} AB = \omega(BA - AB)$

Since $\det(P\bar{P}) = \det P . \det \bar{P}$ is a real number and

$\det \omega(BA - AB) = \omega^n \det(BA - AB)$ and $\det(BA - AB) \neq 0$, then ω^n is a real number. This is possible only when n is divisible by 3.

8. Let (h,k) be the vertex of the graph of f. As the graph of f intersects the x-axis twice, we can assume that $f(x) = a(x-h)^2 + k$ with $\frac{-k}{a} > 0$. Let $s = \sqrt{\frac{-k}{a}}$; then the x-intercepts of the graph of f are $h \pm s$. Because $g(x) = -f(100 - x) = -a(100 - x - h)^2 - k$, it follows that the x-intercepts of the graph of g are $100 - h \pm s$.

The graph of g contains the point (h, k); thus

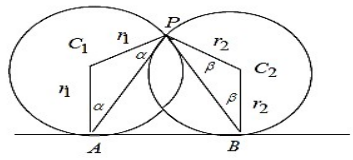
$$k = f(h) = g(h) = -a(100 - 2h)^2 - k,$$

from which $h = 50 \pm \frac{\sqrt{2}}{2}s$. Regardless of the sign in the expression for h, the four x-intercepts in order are

$$50 - s\left(1 + \frac{\sqrt{2}}{2}\right) < 50 - s\left(1 - \frac{\sqrt{2}}{2}\right) < 50 + s\left(1 - \frac{\sqrt{2}}{2}\right) < 50 + s\left(1 + \frac{\sqrt{2}}{2}\right).$$

Because $x_3 - x_2 = 150$, it follows that $150 = s(2 - \sqrt{2})$, that is $s = 150\left(1 + \frac{\sqrt{2}}{2}\right)$. Therefore $x_4 - x_1 = s(2 + \sqrt{2}) = 450 + 300\sqrt{2}$, and then $m - n + p = 450 - 300 + 2 = 152$

9.



ΔPAB, $\frac{2r_1 \cos\alpha}{\sin(90-\beta)} = \frac{2r_2 \cos\beta}{\sin(90-\alpha)} = 2R \Rightarrow R^2 = r_1, r_2 \Rightarrow R = 6$

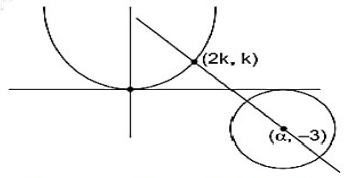
10. $x^2 = 4ky \Rightarrow 2x = 4k\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{x}{2k} \Rightarrow \left(\frac{dy}{dx}\right)_{(2k,k)} = 1$

$\therefore$ Slope of normal at $(2k,k) = -1$

$\therefore$ Equation of normal $y - k = -1(x - 2k) \Rightarrow x + y = 3k$

If $k = 1$, equation of normal $x + y = 3$

$\because$ Shortest distance lies along the common normal



$\therefore$ It passes through $(\alpha, -3); \alpha - 3 = 3 \Rightarrow \alpha = 6$

$\Rightarrow$ *If* $k = 2, x + y = 6$ passes through $(\alpha, -3) \therefore \alpha = 9$

$\Rightarrow$ *If* $\alpha = 20 \therefore x + y = 3k$ passes through $(20, -3)$ $\therefore k = \frac{17}{3}$

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11.

Reflection of (5, 7) w. r. t tangent y = x is (7, 5) line joining (2, 3) & (7, 5) meets y = x at

$$\left(\frac{11}{3},\frac{11}{3}\right) \rightarrow SP + S'P = 2a \Rightarrow a = \frac{\sqrt{29}}{2}$$

$$2ae = 5 \Rightarrow e = \frac{5}{\sqrt{29}} \Rightarrow b = 1$$

12.

The probability that all 5 tosses will be tails is $\frac{1}{2^5} = \frac{1}{32}$.

Otherwise, if at least one of the tosses comes out heads, then at the end the die will be in a random position, because each roll has an equal probability of showing any particular number. The probability that at least one toss is heads is $1 - \frac{1}{32}$, or $\frac{31}{32}$. Given that condition, the probability that the die will be 1 is $\frac{1}{6}$. So the probability that at least one toss is heads and the die ends at 1 is $\frac{31}{32} \cdot \frac{1}{6} = \frac{31}{192}$.

Hence the total probability is $\frac{1}{32} + \frac{31}{192} = \frac{37}{192}$

13.

$$\begin{aligned} 52z_1 - 20z_2 &= 52z_1\frac{z_2\overline{z_2}}{13^2} - 20z_2\frac{z_1\overline{z_1}}{5^2} \\ &= 4z_1z_2\left(\frac{\overline{z_2}}{13} - \frac{\overline{z_1}}{5}\right) \\ &= z_1z_2\left(\frac{20\overline{z_2} - 52\overline{z_1}}{65}\right). \end{aligned}$$

$$\begin{aligned} z_1z_2 &= -65\frac{52z_1 - 20z_2}{52\overline{z_1} - 20\overline{z_2}} \\ &= -65\frac{52z_1 - 20z_2}{\overline{52z_1 - 20z_2}} \\ &= -65\frac{3(4+7i)}{3(4-7i)} \\ &= -65\frac{(4+7i)^2}{4^2+7^2} \\ &= -(4+7i)^2 \\ &= 33 - 56i \end{aligned}$$

14.

$$\Rightarrow x + \frac{x^2}{2} + \frac{x^3}{3} + \ldots\ldots\infty = -\ln(1-x)$$

$$\Rightarrow 1 + \frac{x}{2} + \frac{x^2}{3} + \ldots\ldots\infty = -\frac{\ln(1-x)}{x}$$

$$\Rightarrow \int_0^1 \left(1 + \frac{x}{2} + \frac{x^2}{3} + \ldots\ldots\infty\right)dx = -\int_0^1 \frac{\ln(1-x)}{x}dx$$

$$\Rightarrow -\int_0^1 \frac{\ln x}{1-x} = a \Rightarrow \int_0^1 f(x)dx = -a \Rightarrow I_1 = -a$$

$$I_2 = \int_0^1 \left(\frac{x\ln x}{1-x^2}\right)dx \text{ , let } x^2 = t \Rightarrow 2x\,dx = dt$$

$$I_2 = \frac{1}{2}\int_0^1 \left(\frac{\ln\sqrt{t}}{1-t}\right)dt = \frac{1}{4}\int_0^1 \left(\frac{\ln t}{1-t}\right)dt = \frac{1}{4}I_1 \Rightarrow I_1 - I_2 = \frac{3}{4}I_1 = -\frac{3}{4}a$$

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$$\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r\ln\left(\frac{r}{n}\right)}{n^2\left(1-\frac{r^2}{n^2}\right)}=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}.\frac{r}{n}.\frac{\ln\left(\frac{r}{n}\right)}{1-\left(\frac{r}{n}\right)^2}$$

$$=\int_0^1\frac{x\,dx}{1-x^2}=\frac{1}{4}(-a)=-\frac{a}{4}$$

15.

$$I(a,b)=\left[\cos^a x\ \tfrac{1}{b}\sin bx\right]_0^{\frac{\pi}{2}}-\int_0^{\frac{\pi}{2}}-a\cos^{a-1}x\ \sin x\ \tfrac{1}{b}\sin bx\ dx$$

$$=0+\int_0^{\frac{\pi}{2}}a\cos^{a-1}x\ \sin x\ \frac{1}{b}\sin bx\ dx=\frac{a}{b}\int_0^{\frac{\pi}{2}}\cos^{a-1}x\ \sin x\ \sin bx\ dx$$

$\cos(b-1)x=\cos bx\cos x+\sin bx\sin x$

So $I(a,b)=\frac{a}{b}\int_0^{\frac{\pi}{2}}\cos^{a-1}x\ (\cos(b-1)x-\cos bx\cos x)\,dx$

$=\frac{a}{b}[I(a-1,b-1)-I(a,b)]$

Thus $I(a,b)=\frac{a}{a+b}I(a-1,b-1)$ as required.

(ii) Suppose

$$I(k,k+2m+1)=(-1)^m\frac{2^k\,k!\ (2m)!\ (k+m)!}{m!\ (2k+2m+1)!}$$

Thue s $I(a,b)=\dfrac{a}{a+b}I(a-1,b-1)$ as required

$$I(0,2m+1)=\int_0^{\frac{\pi}{2}}\cos(2m+1)x\ dx=\frac{1}{2m+1}[\sin(2m+1)x]_0^{\frac{\pi}{2}}$$

$=\frac{1}{2m+1}$ if m is even or $=\frac{-1}{2m+1}$ if m is odd , or alternatively $(-1)^m\frac{1}{2m+1}$

$$2I(a,b)=\frac{a}{b}[I(a-1,b-1)-I(a-1,b+1)]$$

16. (P) $f(x)=\int_1^x\sqrt{\frac{t-1}{t+1}}dt;\ puting\ \frac{u}{2}=t$

We have $\int_2^x\sqrt{\frac{u-2}{u+2}}du=2f(x/2)$

(Q) Putting $u+2=2t$ gives $\int_0^x\sqrt{\frac{u}{u+4}}du=2f\left(1+\frac{x}{2}\right)$

(R) Putting $u=3t+2$

(S) Putting $u^2=t$

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17.

$\text{As } \operatorname{sgn}\left(\frac{1}{\{x\}}\right)=1\ \forall x\in R-1$

$\therefore f(x)=\left[\frac{2x}{\pi}\right]\operatorname{sgn}\left(\frac{1}{\{x\}}\right)=\left[\frac{2x}{\pi}\right]\forall x\in R-1$

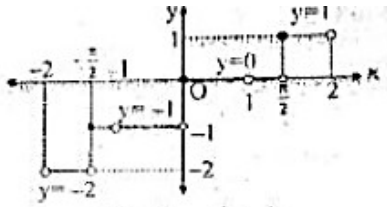
$\text{As } \forall x\in(-2,\ 2)\Rightarrow \frac{-4}{\pi}<\frac{2x}{\pi}<\frac{4}{\pi}$

$\therefore \left[\frac{2x}{\pi}\right]$ is discontinuous when $\frac{2x}{\pi}=-1,\ 0,\ 1$

$\Rightarrow x=\frac{-\pi}{2},\ 0,\ \frac{\pi}{2}$

Clearly $f(x)$ is discontinuous and non-derivable at

$x=\frac{-\pi}{2},\ -1,\ 1,\ \frac{\pi}{2}$ i.e., 5 points.



Graph of $y=\left[\frac{2x}{\pi}\right]\operatorname{sgn}\left(\frac{1}{\{x\}}\right)$ in $(-2,\ 2)$

PHYSICS

18. COM: $mu=3mv_{cm}\Rightarrow v_{cm}=\frac{u}{3}$

COL about CM: $mu\left(\frac{l}{2\sqrt{3}}\right)=3\left[m\left(\frac{l}{\sqrt{3}}\right)^2\right]\omega\Rightarrow\omega=\frac{u}{2\sqrt{3}l}$

After the collision both v_{cm} and ω remain constant and all three point masses will revolve around CM with a constant speed of $\omega\frac{l}{\sqrt{3}}=\frac{u}{6}$, with respect to CM.

Therefore, speed of each mass is $\sqrt{\left(\frac{u}{3}\right)^2+\left(\frac{u}{6}\right)^2+2\left(\frac{u}{3}\right)\left(\frac{u}{6}\right)\cos(\omega t+\delta)}$, where $\delta=\pi,\frac{\pi}{3},-\frac{\pi}{3}$ for A, B, C respectively.

19. Current from $r=0$ to $r=b(\le a)$ is $\int_0^b j(2\pi r)\,dr=\int_0^b\frac{j_o r^2}{a}(2\pi r)\,dr=\frac{\pi j_o b^4}{2a}$

Total current is at $b=a\Rightarrow i=\frac{\pi j_o a^3}{2}$

For half the current, $\frac{\pi j_o b^4}{2a}=\frac{1}{2}\left(\frac{\pi j_o a^3}{2}\right)\Rightarrow b=\frac{a}{\sqrt[4]{2}}$

From Ampere's law: $B(2\pi r)=\begin{cases}\mu_o\int_0^r j(2\pi r)\,dr & \text{if } 0\le r\le a\\ \mu_o i_{total} & \text{if } r>a\end{cases}\Rightarrow B=\begin{cases}\frac{\mu_o j_o r^3}{4a} & \text{if } 0\le r\le a\\ \frac{\mu_o j_o a^3}{4r} & \text{if } r>a\end{cases}$

20. Flux in upper half is same as flux in xz plane cutting the cylinder, it's twice of flux due to each.

$\phi_B=2\int_a^{3a}\frac{\mu_o i}{2\pi r}L\,dr=\frac{\mu_o iL}{\pi}\ln 3$

21. Moment of inertia about H is $I=\left(mR^2+mR^2\right)+2\left(m\left(\sqrt{3}R\right)^2\right)=8mR^2$

Distance of the centre of mass of the system from H is

$d=\frac{mR+m\left(\sqrt{3}R\right)\cos\left(\frac{\pi}{6}\right)+m\left(\sqrt{3}R\right)\cos\left(\frac{\pi}{6}\right)}{3m}=\frac{4R}{3}$

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Therefore, time period $T = 2\pi\sqrt{\frac{I}{(3m)gd}} = 2\pi\sqrt{\frac{8mR^2}{(3m)g\left(\frac{4R}{3}\right)}} == 2\pi\sqrt{\frac{2R}{g}}$

22. The plate P and left side of Q will form a capacitor of capacitance $\frac{\varepsilon_o A}{3d}$, the plate S and right side of R will form a capacitor of capacitance $\frac{\varepsilon_o A}{2d}$, these two are in series with an effective capacitance $\frac{\varepsilon_o A}{5d}$, which is directly connected across the battery. Therefore, a charge of $\frac{\varepsilon_o A}{5d}\varepsilon$ will appear on left of Q and S, whereas a charge of $-\frac{\varepsilon_o A}{5d}\varepsilon$ will appear on P and right of R.

The right side of Q and left side of R will form a capacitor of capacitance $\frac{\varepsilon_o A}{d}$ directly across the battery leading to a charges $+\frac{\varepsilon_o A}{d}$ and $-\frac{\varepsilon_o A}{d}$ on them respectively.

Total charge on Q and R are $\pm\left(\frac{\varepsilon_o A}{d}+\frac{\varepsilon_o A}{5d}\right)=\pm\frac{6\varepsilon_o A}{5d}$ respectively.

23. 300 x 1 x (50 – 30) = Ms(30 – 20)
Ms = 600 cal/g
Water equivalence of calorimeter = 600/1 = 600g
100S(120 – 40) = 400 x 1 x (40 – 20) + Ms(40 – 20)
100S(80) = 400 x 20 + 600 x 20
S = 2.5cal/g-K

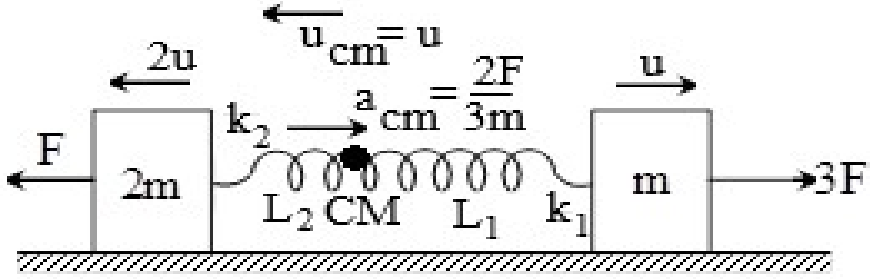
24. O will have minimum tension, say T_o.
Then any point P such that rope length OP = s has a tension T at an angle θ with the horizontal such that:
$T\sin\theta = \lambda sg$
$T\cos\theta = T_o$
$\Rightarrow \tan\theta = \frac{dy}{dx} = \frac{\lambda sg}{T_o}$ and $s = \int\sqrt{(dx)^2+(dy)^2}$

25. Convert all mediums to refractive index 4.2, then
AS, SB will become $\frac{4.2}{1.05}\times 7 = 28$
BP will become $\frac{4.2}{2.1}\times 7 = 14$
f will become $\frac{4.2}{2.1}\times 14 = 28$
$\frac{1}{V}+\frac{1}{-(28+14)}=\frac{-1}{28}$
$V = -3 \times 28 = -84\ cm$
From Eye = $\left|\ 84-(20+28+28+14)\ \right| = 6cm$

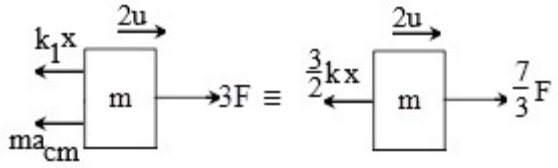
26. At the initial instant the velocity of CM is $\frac{2m(2u)-mu}{2m+m} = u$ to the left and acceleration of CM throughout the journey is $a_{cm} = \frac{3F-F}{2m+m} = \frac{2F}{3m}$ to the right as shown in the figure. CM divides the

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separation between the two blocks as L_1 and L_2, with spring constants k_1 and k_2 such that $L_1+L_2=L$, $mL_1=2mL_2$ and $k_1L_1=k_2L_2=kL$. Therefore, $k_1=\frac{3}{2}k, k_2=3k$.



Now we can shift the frame of reference to CM and study right block.
With respect to CM:



Work-Energy Theorem from initial to max extension:

$$\frac{7}{3}Fx=\left[\frac{1}{2}\left(\frac{3}{2}k\right)x^2+0\right]-\left[0+\frac{1}{2}m(2u)^2\right]\Rightarrow x=0.1$$

In CM frame both blocks will move in the inverse ratio of masses, implies maximum extension in k_2 will be half of x. Therefore, total extension is $1.5x = 0.15 = 15$cm

27. $^{233}_{92}U^{+92}\rightarrow {}^{205}_{82}Pb^{+82}+\alpha\,{}^{4}_{2}He^{+2}+\beta e^{-}$

$233=205+4\alpha\Rightarrow\alpha=7$ and $92=82+2\alpha-\beta\Rightarrow\beta=4$

28. Let the linear mass density of the rod be λ and the steady state acceleration be a then

$F+24=\lambda(4l)a\Rightarrow a=\frac{F+24}{4\lambda l}$, where AB = l

In AB segment, at a distance x from end A:

$$T=\lambda xa\Rightarrow T=\lambda x\left(\frac{F+24}{4\lambda l}\right)\Rightarrow T=\left(\frac{F+24}{4l}\right)x;\forall x\in[0,l)$$

As is positive for all x, implies the complete AB will elongate

In a dx length, we can say: $\frac{T}{A}=Y\frac{dl}{dx}\Rightarrow dl=\frac{Tdx}{YA}$

$$\Rightarrow\Delta(AB)=\int_0^l\left(\frac{F+24}{4lYA}\right)xdx$$

$$\Rightarrow\Delta(AB)=\left(\frac{F+24}{8YA}\right)l\text{ ; Elongation}$$

In BC segment, at a distance x from end C:

$$24-T=\lambda xa\Rightarrow 24-T=\lambda x\left(\frac{F+24}{4\lambda l}\right)\Rightarrow T=24-\left(\frac{F+24}{4l}\right)x;\forall x\in[0,3l)$$

In a dx length, we can say: $\frac{T}{A}=Y\frac{dl}{dx}\Rightarrow dl=\frac{Tdx}{YA}$

$$\Rightarrow\Delta(BC)=\frac{1}{YA}\int_0^{3l}\left[24-\left(\frac{F+24}{4l}\right)x\right]dx=\frac{1}{YA}\left[72l-9l\left(\frac{F+24}{8}\right)\right]$$

$$\text{As }\Delta(AB)+\Delta(BC)=0\Rightarrow\left(\frac{F+24}{8YA}\right)l+\frac{1}{YA}\left[72l-9l\left(\frac{F+24}{8}\right)\right]=0\Rightarrow F=48N$$

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29. Reading = 0.60 + [21 x (1MSD – 1VSD)/1MSD] x 1MSD = 0.60 + [21 x 2/25] x 0.05 = 0.634mm

30. When the key is closed let a charge x circulate in clockwise sense, then

$$\varepsilon-\frac{\frac{\varepsilon C}{4}+x}{2C}-\frac{x}{C}=0\Rightarrow x=\frac{7}{12}\varepsilon C$$

Heat generated is $W + U_i - U_f = \varepsilon\left(\frac{7\varepsilon C}{12}\right)+\frac{\left(\frac{\varepsilon C}{4}\right)^2}{2(2C)}-\left[\frac{\left(\frac{\varepsilon C}{4}+\frac{7\varepsilon C}{12}\right)^2}{2(2C)}+\frac{\left(\frac{7\varepsilon C}{12}\right)^2}{2(C)}\right]=\frac{49}{192}C\varepsilon^2$

31. At the start the friction between them will be kinetic in nature with a magnitude of $\mu mg = 4$, it will act opposite to A's velocity on A.

For A: $a_A = 4, v_A = 2-4t$

For B: $a_B = 9t+4, v_B = \frac{9}{2}t^2+4t$

This will go on till $v_A \geq v_B \Rightarrow 2-4t \geq \frac{9}{2}t^2+4t \Rightarrow \frac{9}{2}t^2+8t-2\leq 0 \Rightarrow t \leq \frac{2}{9}s$

and they reach a speed of $\frac{10}{9}m/s$ at $t=\frac{2}{9}s$

Now, the friction between them will be static and they will move together with an acceleration

$$a=\frac{9t}{2}\Rightarrow\int_{\frac{10}{9}}^{v}dv=\int_{\frac{2}{9}}^{t}\frac{9t}{2}dt\Rightarrow v=\frac{9}{4}t^2+1$$

They will move together till friction reaches the limiting value at $a=\frac{9t}{2}=4\Rightarrow t=\frac{8}{9}s$

and they reach a speed of $\frac{25}{9}m/s$ at $t=\frac{8}{9}s$

After $t=\frac{8}{9}s$, friction is kinetic again but in the direction of velocity of A on A, implies

For A: $a_A = 4, v_A = \frac{25}{9}+4\left(t-\frac{8}{9}\right)=4t-\frac{7}{9}$

For B: $a_B = 9t-4\Rightarrow\int_{\frac{25}{9}}^{v}dv=\int_{\frac{8}{9}}^{t}(9t-4)dt\Rightarrow v=\frac{9}{2}t^2-4t+\frac{25}{9}$

32. $dA=-\lambda A dt\Rightarrow A=2Ne^{-\lambda t}$

$dB=\lambda A dt-2\lambda B dt\Rightarrow dB=2\lambda Ne^{-\lambda t}dt-2\lambda B dt$

$\Rightarrow dB+2\lambda B dt=2\lambda Ne^{-\lambda t}dt\Rightarrow d\left(Be^{2\lambda t}\right)=2\lambda Ne^{\lambda t}$

$\Rightarrow B=Ne^{-2\lambda t}+2N\left(e^{-\lambda t}-e^{-2\lambda t}\right)\Rightarrow B=2Ne^{-\lambda t}-Ne^{-2\lambda t}$

33. $\phi_{2,1}=\frac{2\pi}{\lambda}\left[\mu(2a)\times 2+(2\mu)(3a)\times 2\right]+\pi$

The extra π at the end is because light reflected at A is a reflection at the boundary of rarer to denser medium.

$\Rightarrow \phi_{2,1}=\frac{2\pi}{\lambda}[16\mu a]+\pi\Rightarrow\phi_{2,1}=\frac{32\pi}{\lambda}\mu a+\pi$

For constructive interference $\phi_{2,1}=2n\pi\Rightarrow\frac{32\pi}{\lambda}\mu a+\pi=2n\pi\Rightarrow\frac{\mu a}{\lambda}=\frac{(2n-1)}{32}\Rightarrow\left(\frac{\mu a}{\lambda}\right)_{\min}=\frac{1}{32}$

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For destructive interference $\phi_{2,1}=(2n+1)\pi \Rightarrow \frac{32\pi}{\lambda}\mu a+\pi=(2n+1)\pi \Rightarrow \frac{\mu a}{\lambda}=\frac{n}{16} \Rightarrow \left(\frac{\mu a}{\lambda}\right)_{\min}=\frac{1}{16}$

Now, $\phi_{3,1}=\frac{2\pi}{\lambda}\left[\mu(2a)\times 8+(2\mu)(3a)\times 4\right]$

No extra π at the end because the lights reflected at A, I, K and N all surfer an extra π phase change as they are reflections at the boundary of rarer to denser medium. Therefore, the net change in extra phase difference between the two light is 3π – π = 2π, which is equivalent to zero extra phase difference.

$\Rightarrow \phi_{3,1}=\frac{80\pi}{\lambda}\mu a$

For constructive interference $\phi_{3,1}=2n\pi \Rightarrow \frac{80\pi}{\lambda}\mu a=2n\pi \Rightarrow \frac{\mu a}{\lambda}=\frac{n}{40} \Rightarrow \left(\frac{\mu a}{\lambda}\right)_{\min}=\frac{1}{40}$

For destructive interference $\phi_{3,1}=(2n+1)\pi \Rightarrow \frac{80\pi}{\lambda}\mu a=(2n+1)\pi \Rightarrow \frac{\mu a}{\lambda}=\frac{(2n+1)}{80} \Rightarrow \left(\frac{\mu a}{\lambda}\right)_{\min}=\frac{1}{80}$

34. If $a \geq 5$, the highest point of the trajectory is the top most point of the circle.

$T_{top}+mg=\frac{mv_{top}^2}{R}$ and $\frac{1}{2}m\left(\sqrt{agR}\right)^2=\frac{1}{2}mv_{top}^2+mg(2R)$

$\Rightarrow \frac{T_{top}}{mg}=a-5$

If $2 \leq a < 5$, circular motion ceases and parabolic motion starts. At the highest point of the trajectory the rope will be slack and tension will be zero.

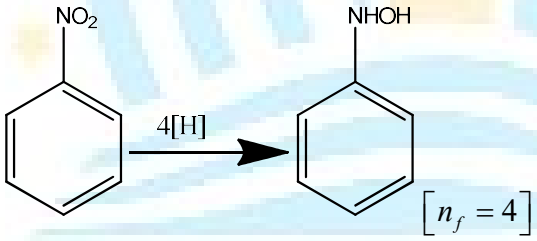
If $0 < a < 2$, the particle swings like a pendulum with an amplitude of β, such that

$T_{top}-mg\cos\beta=0$ and $\frac{1}{2}m\left(\sqrt{agR}\right)^2=mg(R-R\cos\beta)$

$\Rightarrow \frac{T_{top}}{mg}=1-\frac{a}{2}$

CHEMISTRY

35.



$[n_f=4]$

Charge $=\frac{100}{1000}\times\frac{6\times 60}{96500}F$

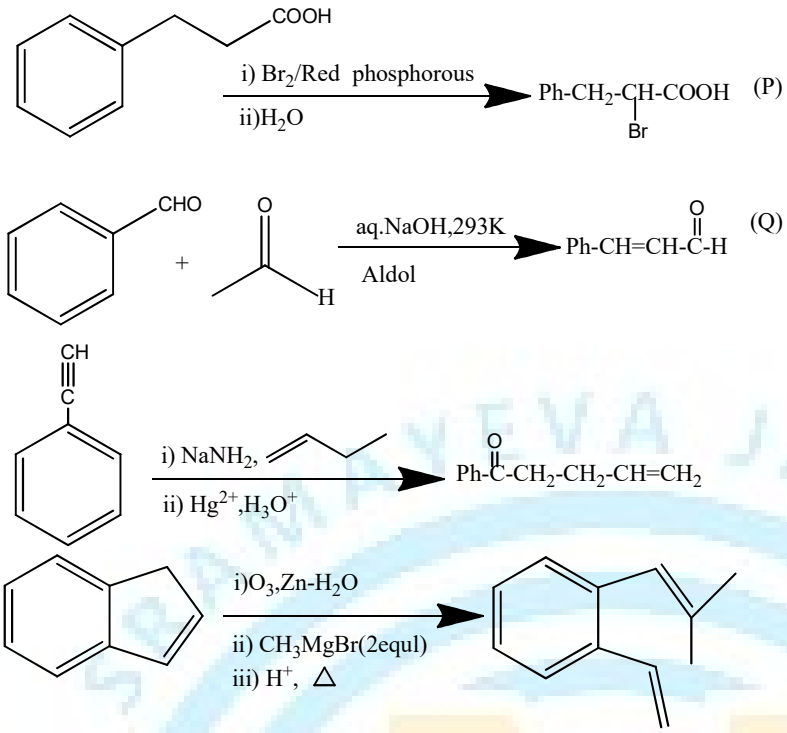
Moles of nitro benzene $=\frac{100}{1000}\times\frac{6\times 60}{96500}\times\frac{1}{4}$

$W_{nitrobenzene}=\frac{100}{1000}\times\frac{6\times 60}{96500}\times\frac{123}{4}=0.01147$

$\%\frac{w}{W}=\frac{0.01147}{0.2}\times 100=5.73\%$

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36.



37.

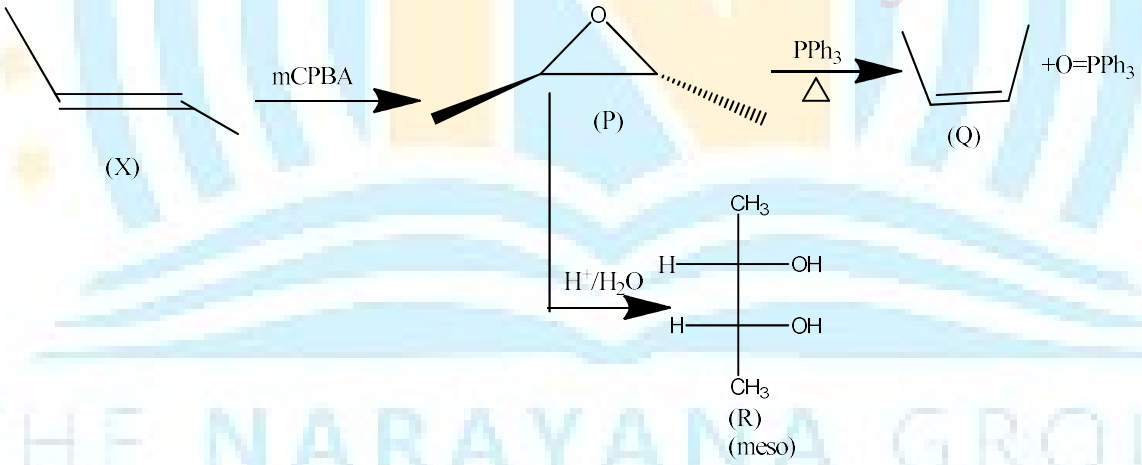
$'P'\ is\ K_3\left[Cu(CN)_4\right] \Rightarrow \mu = 0$

$"Q"\ is\ K_2\left[CuCl_4\right] \Rightarrow \mu = 1.73B.M$

$"S"\ is\ \left[Fe(H_2O)_5(NO)\right]SO_4 \Rightarrow \mu = 3.87B.M$

$"V"\ is\ HMnO_4 \Rightarrow \mu = 0$

38.



39. $FeO + Fe_2O_3 + inert\ impurity = 5g$

$Fe_2O_3 + 2KI \rightarrow \underset{(n_f=2)}{I_2} + Fe^{T2}$

$\underset{(n_f=2)}{I_2} + 2Na_2S_2O_3 \rightarrow Na_2S_4O_6 + NaI$

No. of equivalent of Fe_2O_3 = No. of equivalents of KI=No. of equivalents of I_2

$\Rightarrow$ Moles of $Fe_2O_3 = \dfrac{0.5\times10^{-2}}{2} = \dfrac{5\times10^{-3}}{2} = 0.25\times10^{-3}$ [in 40 mL solution]

So total moles of $Fe_2O_3 = 2.5\times10^{-3}\times5 = 12.5\times10^{-3}$

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So, moles of Fe_2O_3 = Moles of $FeO = 0.0125$

So mass of $Fe_2O_3 = 0.0125 \times 160 = 2g$

$\Rightarrow \%\ \ of\ Fe_2O_3 = 40\%$

$Mass\ of\ FeO = 0.0125 \times 72 = 0.9g$

$\Rightarrow \%\ of\ FeO = 18\%$

So, $\%$ of inert impurity $= = 100 - 40 - 18 = 42\%$

40. $\dfrac{[B]}{[C]} = \dfrac{3K_1}{8K_1} = \dfrac{3}{8} = \alpha$

$\dfrac{[C]}{[D]} = \dfrac{8K_1}{7.5K_1} = \dfrac{8}{7.5} = \dfrac{16}{15} = \beta$

41. Ploymerisation of caprolectum is nylon-6

42. CONCEPTUAL

43.

$$Z = \frac{1}{\sqrt{2}}\pi\sigma^2\bar{U}\left[\frac{N}{V}\right]^2$$

$$\lambda = \frac{1}{\sqrt{2}\pi\sigma^2}\left[\frac{V}{N}\right]$$

N=number of gas molecules

σ = collision area (depends on molecular size only)

U_{avg} = average velocity of gas molecules $= \sqrt{\dfrac{8RT}{\pi M}}$

Hence correct statements are (ii) and (vii)

44. $X \rightarrow Na_4\left[Fe(CN)_5(NOS)\right]$

$CN^- \rightarrow$ ambidentate with donor sites C and N

$NOS^- \rightarrow$ ambidentate with donor sites N,O,S

45. $\alpha -$ methyl carbonyl compounds & $\beta -$ methyl alcohols will give iodo from test.

[II,IV,V,VI]

46. $K_XH(C_2O_4)_Y + KMnO_4 \rightarrow CO_2 + Mn^{+2}$

n –factor of $K_XH[C_2O_4]_Y \Rightarrow 2Y$

&also $2Y = X + 1$

m.eq of salt+meq of mohr salt =m.eq of $KMnO_4$

on solving , $Y = 2$, since $X + 1 = 2Y \therefore X = 3$

Molecular formula of the compound is $K_3H[C_2O_4]_2$

47. Initial moles $= M \times V = 5 \times 1.5 = 7.5mol$

No of moles consumed =1.5 mol

No. of moles of H_2SO_4 remaining $= 7.5 - 1.5 = 6$

Final molarity $= \dfrac{6}{1.5} = 4M$

48. CONCEPTUAL

49. CONCEPTUAL

50. Ozonolysis &aldol reaction

51. Electrophillic aromatic substitution (EAS) reaction

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